Can Rooftop Solar Reduce Manila's Energy Dependence?

Problem, findings, proposed solution and Tableau Public dashboard plan

Executive summary

The Philippines remains fossil-fuel dependent, relies heavily on Middle East oil, and Meralco customers carry rising generation costs and pass-through charges. We analysed a building-level rooftop solar dataset of 301,381 Manila rooftops to test whether solar can ease each of these three problems.

Question	Verdict	Why
Reduces fossil fuel dependence?	**Supported, partially**	Full potential is about 6.4 TWh/yr, roughly 9% of coal generation. Realistic adoption (25%) is about 2%. Daytime only without storage.
Reduces Middle East dependence?	**Weak, indirect**	Oil is only about 1% of power generation. The exposure is in transport and industrial fuel, which rooftop solar reaches only through electrification or diesel-genset replacement.
Reduces dependence on Meralco?	**Supported at customer level**	Self-use avoids the generation charge (about 64% of the bill) and system loss. Customers stay grid-connected for night demand and distribution.

Key insight: the potential is concentrated. The largest 10% of roofs hold half of all energy potential, so targeting a few thousand large commercial, industrial, public and multi-family roofs captures most of the value.

Read with care: results are technical potential from modelled data. Context figures for national generation, Meralco sales, tariff and emission factor are unverified placeholders and must be checked against DOE, ERC and Meralco sources before publication.

1. The problem

Three linked problems motivate the project:

- **Fossil-fuel dependence.** Coal supplied about 62% of Philippine power generation in 2024 (IEA/Ember figures reported in the press), and more than 80% of that coal is imported. The government targets 35% renewables by 2030 and 50% by 2040.
- **Middle East dependence.** Reports put the share of crude oil imports from the Middle East at 90-98%. The 2026 regional conflict triggered a fuel-supply scare and a declared energy emergency. Oil is only about 1% of power generation, so this problem sits mostly in transport and industry.
- **Meralco cost pass-through.** Generation is roughly 64% of the bill, followed by distribution (12%), transmission (8%), system loss (5%) and other charges (11%). Fuel costs and a weaker peso drove generation-charge increases in 2026, and under-recoveries are being collected over three years. The specific claim that Meralco's own losses are passed to consumers was not fully verified beyond system loss and under-recovery pass-throughs.

Hypotheses tested: (1) rooftop solar can lessen fossil-fuel dependence, (2) it can lessen Middle East dependence, (3) it can lessen dependence on Meralco.

Sources: Power Philippines (IEA coal report), Gulf News (coal vs oil in power), Manila Bulletin and GMA (2026 oil supply), Philstar and GMA (Meralco rates and bill breakdown).

2. The dataset

The source file contains 301,381 rows, one per Manila rooftop. It appears to be modelled rooftop PV output with no documentation, so units are assumed: area in m², capacity in kWp, energy in kWh/yr, and the 'capacity factor' column is really specific yield in kWh/kWp/yr.

Field	Meaning	Note
Surface_area, Potential_installable_area	Roof area and usable area	Median 79 and 43 m²
Peak_installable_capacity	Panel capacity (kWp)	Median 8.4 kWp
Energy_potential_per_year	Annual output (kWh)	Median about 10,300 kWh
Assumed_building_type	5 categories	Multi-family, single-family, industrial, commercial, public
Estimated_tilt, Estimated_building_height	Roof tilt, height	Height had negative and extreme values
Unit_installation_price, Comment	Empty	No cost data, so no payback possible

Cleaning

- Dropped 2 empty columns and 5,397 zero-capacity rows (no installable area). 295,984 rows (98.2%) remain.
- Set 1,951 invalid or missing heights to the building-type median and flagged them. Height does not affect energy results.
- Flagged 14,748 statistical outliers and 296 top-0.1% roofs but kept them, because they are real large roofs.
- Found capacity density constant at 0.19 kWp/m² and energy exactly equal to capacity times yield in every row. This confirms the data is modelled, so correlations between area, capacity and energy are by construction.

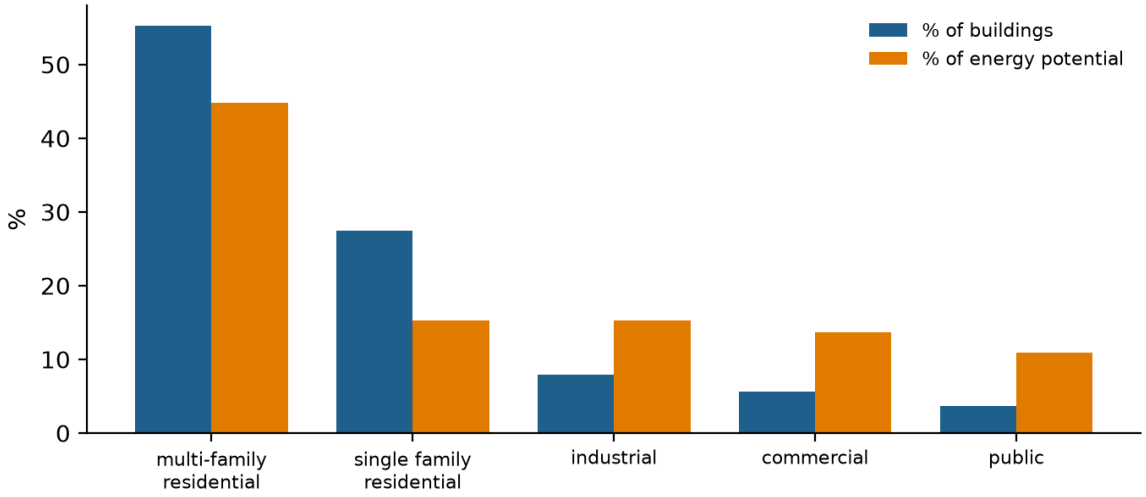
3. Basic findings and analysis

Scale of the resource

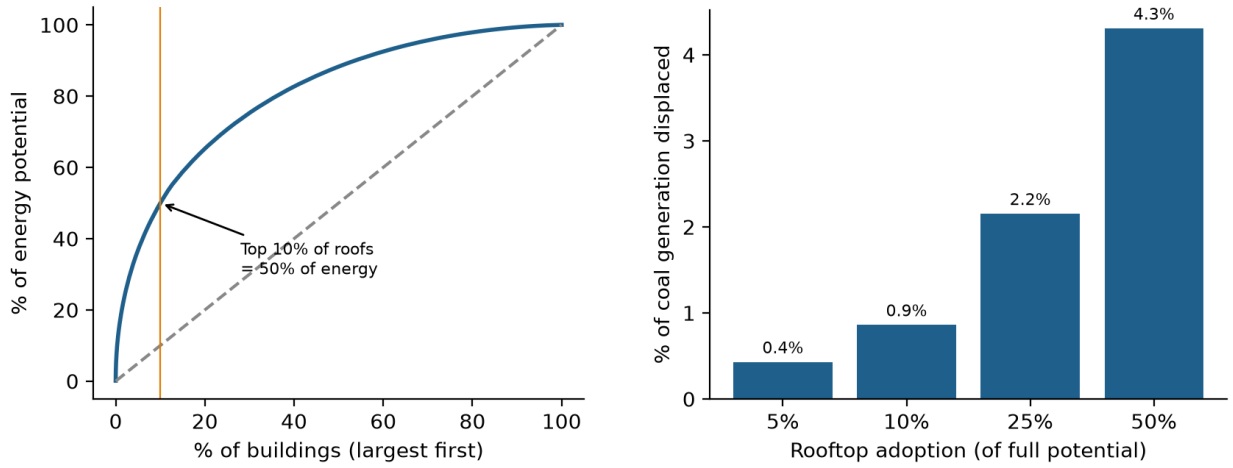
Metric	Result
Buildings analysed	295,984
Usable roof area	26.7 km²
Installable capacity	5.19 GWp
Annual energy potential	6.41 TWh
Fleet capacity factor	14.1% (specific yield about 1,236 kWh/kWp/yr)
Median roof	8.6 kWp, about 10.5 MWh/yr

Using placeholder context values (120 TWh national generation, 62% coal, 50 TWh Meralco sales), this is about 5.3% of national generation, 12.8% of Meralco sales and 8.6% of coal generation.

Where the potential sits



- **Multi-family residential** is the largest block (about 55% of buildings, 45% of energy), but roofs are shared among many households, which complicates ownership and benefit sharing.
- **Commercial, industrial and public** are about 17% of buildings but about 40% of energy (2.56 TWh), with daytime demand and a single decision-maker. They are the practical first target.
- **Single-family** roofs are easy to decide on but small.



Left: concentration curve. About 6,500 roofs (2%) hold 25% of potential; about 30,000 (10%) hold 50%. Right: share of coal generation displaced at different adoption levels (uses the unverified 120 TWh and 62% inputs).

Verdict on the three hypotheses

Hypothesis	Evidence	Verdict
1. Lessens fossil fuel dependence	Full potential displaces about 9% of coal generation and avoids about 4.5 Mt CO_2/yr. At 25% adoption: about 2% of coal and 1.3% of national generation. Evening peak still needs coal/gas without batteries.	Supported, partially
2. Lessens Middle East dependence	Oil is about 1% of power, so at most about 1.2 TWh of oil-fired power is displaceable. The exposure is transport and industrial fuel, outside this dataset.	Weak, indirect only
3. Lessens Meralco dependence	Avoided grid purchases about PHP 61.6 bn/yr at full potential (PHP 15 bn at 25%), assuming PHP 12/kWh and 80% self-use. Customers remain connected for night demand and distribution.	Supported per customer

Caveats

- Technical potential only: no shading, structural, ownership or grid-hosting constraints.
- No installation prices, so no payback, LCOE or NPV.
- No demand data: the 80% self-consumption figure is an assumption, and solar without storage covers daytime load only.
- Summed roof area (about 41 km²) is close to the land area of Manila city, so the dataset's geographic coverage needs confirming.
- National generation, Meralco sales, tariff and emission factor are unverified placeholders.

4. Proposed solution

Rooftop solar is a real but partial answer. The data supports a targeted, staged approach rather than blanket deployment.

Priority	Action	Rationale
1	Target large commercial, industrial and public roofs first (about 17% of buildings, about 40% of energy).	Daytime demand, single owner, fastest self-use and savings.
2	Aggregate the top roofs (about 6,500 roofs give 25% of potential; about 30,000 give 50%).	Highest return per transaction; most of the value in few sites.

Priority	Action	Rationale
3	Design models for multi-family roofs (shared ownership, community solar, net-metering allocation).	Largest block of potential but hardest governance.
4	Pair solar with batteries on the evening peak.	Without storage the coal displacement is capped by daytime load.
5	Link to electrified transport and diesel-genset replacement.	The only credible path to reducing Middle East oil exposure.
6	Use net metering for exports (ERC Resolution No. 09, Series of 2013; verify current rules).	Reduces customers' Meralco purchases; rules need confirmation.

Next analytical steps: add an installation price assumption for payback by building type; add demand data to replace the self-consumption assumption; verify national and Meralco inputs against DOE and ERC sources; confirm the dataset's geographic coverage.

5. Tableau Public dashboard plan

Source: manila_rooftop_solar_clean.csv (about 296k rows, within Tableau Public limits, no map possible because the data has no coordinates). Frame the dashboard as one question: *how much can Manila's rooftops do, where is it, and does it answer our three problems?*

Dashboard contents

Section	Views	Purpose
1. Headline KPIs	Buildings; GWp; TWh/yr; capacity factor; % of coal generation; % of Meralco sales	Scale at a glance. Last two driven by editable parameters.
2. Where the potential is	Energy by building type (buildings share vs energy share); Pareto concentration curve; roof size classes; log-binned capacity histogram	Shows the top 10% of roofs hold 50% of energy.
3. Three-hypothesis panel	Coal displaced gauge; oil about 1% of power chart with indirect-link note; % of Meralco sales and PHP avoided	Directly answers each problem with a verdict.
4. Scenarios	Energy, CO_2 and PHP saved at 5/10/25/50% adoption; derating vs self-consumption heatmap	Shows sensitivity of results.
5. Method footer	Rows cleaned, modelled-data note, technical-potential note, geography caveat	Honest limitations.

Interactivity

- **Parameters:** adoption slider (0-100%), national generation, coal share, Meralco sales, tariff, emission factor, self-consumption. Label them 'assumption, editable'.
- **Filters:** building type and size class, applied across all sheets.
- **Actions:** clicking a building type highlights it everywhere.

Calculated fields

Field	Formula
Capacity factor %	SUM([energy_kwh_yr]) / SUM([capacity_kwp]) / 8760
Energy at adoption	SUM([energy_kwh_yr]) * [Adoption]
% of coal generation	[Energy at adoption] / ([National Gen TWh] * 1e9 * [Coal Share])
Avoided PHP	[Energy at adoption] * [Self Use] * [Tariff]
Cumulative % energy	RUNNING_SUM(SUM([energy_kwh_yr])) / TOTAL(SUM([energy_kwh_yr])), sorted by capacity descending

Layout and tips

- **Top:** title, one-line takeaway, KPI tiles. **Middle left:** building type bars and Pareto curve. **Middle right:** the three verdict tiles. **Bottom:** scenario and size-class charts. **Sidebar:** parameters and filters.
- Use extracts (.hyper), which Tableau Public requires. Data on Tableau Public is publicly visible; this dataset has only building UUIDs.
- Limit the dashboard to six to eight views and use one consistent colour per building type.
- Leave out the correlation heatmap and scatter plots: density is constant, so they carry little information.
- For performance, pre-aggregate by building type and size class, or bin capacity into deciles for the Pareto view.